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Ecospacing as post-growth research frame? Spatial potentials for interspecies wellbeing and regeneration

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ABSTRACT: Within biology, biodiversity is understood as an affluence, a surplus that goes beyond what humans need for functional survival. Yet, like the aesthetic dimension of architecture and the notion of human well-being in sociology, we hypothesize that it is vital in rebalancing our ecological ability of interspecies co-existence and creative impact on the planet. This paper outlines and discusses the process of developing a transdisciplinary post-growth research frame inquiring if interspecies well-being and refraining from excessive consumption and building can be correlated to certain spatial traits in terrestrial and built environments. The ambition is to obtain high quality empirical data that can inform transdisciplinary and transformative action-based research practices while testing whether a co-created local environment enhanced by interspecies well-being holds macros-scale regenerative potentials? This is pursued by systematically mirroring and integrating the notion of Ecospace, describing the biodiversity potential of local terrestrial biotopes in a macro-scale perspective, to embody also the built environment and the human species in a techno- and biosphere continuum.

1 INTRODUCTION

Construction impacts the climate and biodiversity crisis on a colossal scale (UNEP 2023) and influences all spheres expressing human expansion and success and surpassing in volume that of biomass (Elhacham 2020). When determining the environmental impact, using the IPAT equation (Hauschild 2018), T (technological development) is the main focus for the building industry's endeavors towards sustainability. The 'A' (affluence), on the other hand, is a complex notion pointing to the entangled question of whether there are 'places' or situations where abstaining from over-consumption and excessive building would hold also experiences of affluences for lifeforms including humans? Economy of wellbeing, theories of postgrowth and degrowth (Meadows et al. 2004, Jackson 2017) have for decades inquired correlations between wellbeing and economic prosperity without reaching decisive conclusions on local ways of inhabiting neither built nor terrestrial environments. Consequently, leading post-growth theorists call for specific research frameworks and case-studies for the built environment in a recently published extensive review paper (Kallis et al. 2025, p. e71). Answering to this call, this paper explores the development of a transdisciplinary post-growth research framework, by an interdisciplinary group of researchers consisting of architects, biologists and sociologists integrating theories and methods across the fields of architecture, biology and sociology towards identifying local spatial indicators of affluence with low environmental impact; Sufficiency. In this matter, the paper builds upon initial steps taken by the group,

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through which we have identified a conceptual potential to integrate emerging ecological research methods within each of our fields allowing us to juxtapose quantitative and qualitative measures across the three disciplines (Hvejsel et al. 2024). In this paper, this potential is substantiated and specified by studying the correlations between urban sociology and key notions of ecology originating from biology.

This paper's result is a transparent description of a process, where a new transdisciplinary research framework is the aim, insisting on 'communication' with the agency of a living world, while being rooted in disciplinary knowledge across architecture, sociology and biology. Ecospace is a new research framework in itself, in biology, where the biodiversity of a given biotope is qualified through ecological processes (Brunbjerg et al. 2017). In sociology, neighborhoods are qualified through social processes, identified in the level and composition of collective efficacy. In both cases, the environment is a spatial actor inseparable from the species that inhabit it, whether terrestrial or constructed. Hence, this paper describes a segment of a development process towards a research framework that can deliver empirical data from case studies as called for by Kallis et al. (2025), whereby possible spatial correlations between interspecies well-being and sufficiency within planetary boundaries can be identified.

2 DEVELOPING A TRANSDISCIPLINARY POST-GROWTH RESEARCH FRAME

2.1 *Background – Meta-methods for interdisciplinary and transdisciplinary research*

Over the course of a year of semi-weekly interviews between members of the research group, it was possible to identify the field of ecology as the focal point for the collaboration. This was motivated by the observation that in biological theories of Ecospace (Brunbjerg et al. 2017) a well-established correlation between terrestrial biotopes and their specific life forms is demonstrated leaving agency to the living while having an interface to the built environment. Ecospace, describing the biodiversity potential of a local terrestrial biotope, is rooted in community ecology, where shortly after the formulation of ecology as a discipline in 1866 by Ernst Haeckel, Danish biologist Eugen Warming established the fields of plant ecology and community ecology (Watts et al. 2019). The concept of community ecology implied a strong notion of interdependence between species, material resources and life, and this perspective inspired the thinking of the Chicago School of urban sociology, however, failing to document environmental impact and influence architectural practice locally (Angelo & Greenberg 2023). Addressing this knowledge gap, our mutual interdisciplinary interviews led to the identification of a potential to instrumentalize Ecospace as a traveling concept (Neumann & Nünning 2012) that we as architects and sociologists 'mirrored' into the built environment. Using Ecospace as a starting point challenged the inherent anthropocentric approach in sociology and architecture. By imagining 'mirroring' and integration of the terrestrial definition of Ecospace into the built environment, we uncovered a series of principles for a conceptual research design opening a potential to place humans and nonhumans in a continuous Ecospace, opening a direction to identify spatial indicators for regeneration (Hvejsel et al. 2024). In this process sociologists and architects sought to establish an interdisciplinary interface with the biological Ecospace theory together with the authors of the theory. This interface is outlined and presented as the main body of this paper, providing a foundation to formulate common research questions and as an outline for how future case-studies could provide an interdisciplinary empirical ground to progress from Ecospace to Ecospacing, inviting a larger transdisciplinary research community and real-world actors to participate.

2.1.1 *Ecospace theory; a short introduction from Brunbjerg et al. 2017*

A terrestrial biotope is the geospatial basis for Ecospace (see Figure 1), i.e. the physical area in which a given set of organisms live. Every terrestrial biotope - regardless of its ecological setting and spatiotemporal extent - can be described in terms of its Ecospace. Ecospace *position – the first Ecospace component:* reflects the position of the biotope in a (hyper-)space spanned by multiple edaphic, physical or chemical gradients, measured in the biotope such as soil pH,

nutrient availability, soil texture, soil moisture and temperature. Ecospace *expansion*: The second component of Ecospace is defined as the accumulation, formation, and diversification of resources i.e. organic structures and substrates for species to live on and from. *Expansion* consists of pools and structures of organic matter, ultimately derived from the growth of plants and other primary producers, and provides food, shelter and space for other species. Consequently, *expansion* includes both the accumulation and spatial distribution of biomass (structural complexity of vegetation) and the partitioning of biomass into functionally separated carbon pools and structures (diversification of organic matter). Plant species richness, foliage, flowers (providing nectar and pollen), fruits/seeds, tree trunks, branches, dead wood, litter, carcasses and dung, all contribute to the expansion of Ecospace. Ecospace *continuity:* refers to the extension in time and space of a given Ecospace position and expansion. Spatial and temporal continuity are estimated with respect to the biotope in question but are measured for the surrounding area (spatial) and the conditions preceding (temporal) the current Ecospace. Spatial and temporal continuity affect the colonization of a biotope (cf. meta-community theory, Leibold et al. 2004). *Continuity* in Ecospace position enhances the probability of colonization and establishment of species from the species pool and decreases the risk of local extinction. With increasing continuity, the realized species richness in the biotope approaches the potential species richness constrained by the regional species pool and the available conditions and resources. Ecospace *uniqueness*: Given that Ecospace can predict species composition, the contribution of a biotope to β-diversity (the ratio between regional and local species diversity) can be predicted by its uniqueness in position, expansion and continuity in the context of a larger geographic area. *Uniqueness* is not part of the formulation of Ecospace but should be regarded as an extrapolation of Ecospace that enables us to assess the contribution of the biotope to the biodiversity of the surrounding landscape. *Uniqueness* is an estimate of the 'rarity' of the biotope at a larger scale estimated by 1/log (% national cover of Ecospace type) of the given biotope. The higher the uniqueness, the more the biotope is expected to contribute to diversity at larger scales (Brunbjerg et al. 2017).

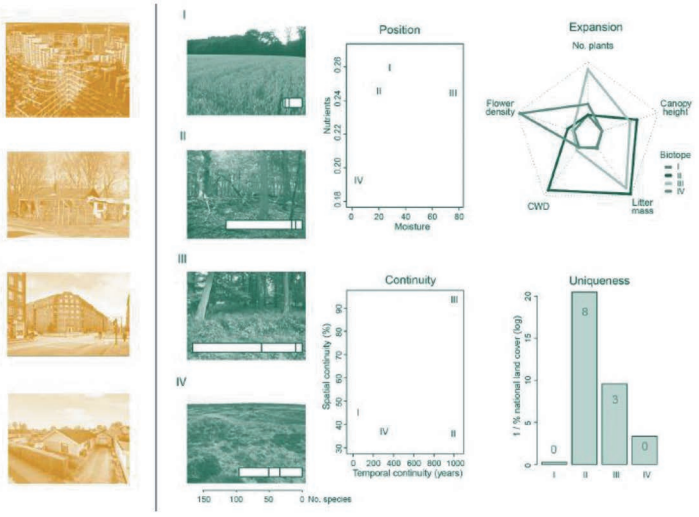


Figure 1. Left: Mirroring Ecospace into the built environment into the 4 Ecospace categories. Right (I-IV): Ecospace for four contrasted biotopes (I-IV) in Denmark, showing how biotopes may vary independently in position, expansion continuity, and uniqueness; and how that affects α-diversity (horizontal white bars). *Position* is represented here by two important gradients: soil moisture (dry to wet) and nutrients (nutrient poor to nutrient rich). *Expansion* is represented by flower density (range = 0-70/m^2), number of vascular plant species (No. plants; range = 0-60), canopy height (range = 0-8 m), litter mass (range = 0-150 g) and coarse woody debris (CWD; range = 0-50 m^3). *Continuity* is divided into spatial (% cover of the Ecospace type within 500 m) and temporal continuity of the biotope (years since last major environmental change). *Uniqueness* is represented by the number of vulnerable or threatened species found in the four biotopes. Photos: Lars Skipper, Copyrights: Held by the authors, reproduced from (Brunbjerg et al. 2017).

2.1.2 *Interfaces of urban sociology and terrestrial ecology*

Within urban sociology the ecological inspiration that drove the Chicago School resulted in the development of the so-called Human Ecology perceiving ecological community as a precursor of society marked by an unreflective form of community (Park 1952. At this foundational stage, interaction is not conscious or reflective but rather dictated by competition for collective survival. However, as society evolves, deliberate and conscious forms of interaction emerge. The instinct-driven layer does not disappear but instead becomes overshadowed by more complex social structures. Society has a dual nature: it is both competitive and cooperative. Individuals act independently, competing for survival, yet they are also deeply connected through shared emotions, traditions, and collective goals. According to Park, the way humans communicate socially—mutually and consciously—develops through different phases and takes various forms: an economic phase marked by conflict, a political phase characterized by adaptation, and a moral phase defined by assimilation. These phases manifest in different disciplines: human geographers study competition within ecological communities, economists examine monetary mediation in conflicts, political scientists analyze institutional adaptation, and sociologists focus on the moral phase, where human interactions transcend political and economic constraints. In this sense, sociology prioritizes interaction based on norms and values (Jørgensen 2018).

Neither biological nor social systems remain static; both experience periodic disruptions that unsettle their equilibrium (Macarthur & Wilson 1967). These disturbances release suppressed forces, triggering rapid and intense change. However, over time, systems adapt, reorganize, and reestablish stability. This cyclical process of disruption, adaptation, and stabilization continuously shapes both terrestrial and social environments. In the urban context, different neighborhoods represent varying stages along the spectrum between instinct-driven communities and those characterized by shared values and moral cohesion. Park and Burgess (1925) introduced key concepts such as "invasion," "succession," "dominance," and the "zone in transition" to describe demographic shifts and social demarcations within the city. According to their framework, neighborhoods oscillate between destabilized and organized communities, influenced by patterns of demographic change. Urban spaces are shaped by local social groups, evolving at different rates and reflecting unique interactions with their environment. Understanding these dynamics offers valuable insights into urban life and transformation. Building on classical Human Ecology, Robert Sampson introduced the concept of Collective Efficacy to highlight the importance of understanding neighborhood variations through local routines, activity patterns, organizational structures, social networks, segregation, and resource distribution (Sampson 2011). Collective efficacy comprises four key elements (Sampson 2012):

1. Ecology of Routine Activities (Behavior Settings) – The everyday patterns of action within social environments that shape interactions and behaviors.
2. Organizations (Civic Structure) – The formal and informal networks that structure community life and encourage civic participation.
3. Density of Social Ties/Social Networks (Exchange) – The strength of interpersonal connections and the exchange of social and material resources within these networks.
4. Segregation (Resource Stratification) – The separation of groups and unequal access to resources, often defined by social or economic divisions.

The term neighborhood here carries a dual meaning: it refers both to a physical geographic space and to a social construct that encompasses shared experiences and interactions. According to Sampson (2012), there is no definitive theoretical or empirical answer to what constitutes a neighborhood. Traditionally, the concept has been associated with shared values and localized social bonds, but neighborhoods are dynamic, evolving through stages of trust, interaction, and identity formation. Consequently, place cannot be reduced to rigid administrative boundaries, which are often arbitrary (Jakobsen et al. 2022; Lund 2019; Lund & Jørgensen 2024). Instead, place is shaped by local social relations, demographic patterns, and economic factors. Collective efficacy thus reflects the potential for collective action, extending beyond formal participation in political or civic organizations. It varies across different localities, manifesting through mutual trust and shared expectations among residents to address local challenges together. A neighborhood's social capacity—the ability to sustain local social life and engage residents in

communal activities—is central to collective efficacy and plays a crucial role in fostering biodiversity and a local green transition through reciprocity, direct social interactions, and shared responsibility. Hence, the above study of interfaces between urban sociology and terrestrial ecology points to a set of parallels between the 'ecological processes' that qualify the biodiversity of a given terrestrial biotope and the 'social processes' that qualify the capacity for life of the urban environments; in both cases processes which are inseparable from the geospatial environmental conditions which they inhabit, whether terrestrial or constructed. These parallels provide an entrance to zoom in on these spatial conditions as means to mirror and integrate the notion of Ecospace, describing the biodiversity potential of local terrestrial biotopes, to embody the built environment and the human species towards situation and activation of the regenerative potential of local biotopes in techno- and biosphere continuum (Lyle, 1994).

2.3 *Ecospace mirrored towards integration into the realm of the built environment.*

When attempting to integrate the built environment into the Ecospace theory as proposed, where the living realms of Ecospace are construed in light of notions from the built environment, we systematically 'mirror' the components and variables of the Ecospace theory one by one, to echo each phenomena in the built environment.

Ecospace *position (built)*: The built environment perceived as buildings made of concrete, brick, stone, wood, metal and glass affects a biotope's soil pH, nutrient availability, soil texture, soil moisture and temperature. Soil pH is influenced by soil compaction from building construction and maintenance processes and the drainage strategy chosen for roofs and ground coverings. These things also affect nutrient availability, soil texture and soil moisture. Buildings accumulate cold and heat in their thermal mass and shade the terrain, which also affects temperature and other soil conditions. This happens on surrounding areas, on facades and roofs, and in transition zones between indoors and outdoors. There is turbulence or lee created by buildings, affecting the wind conditions. Buildings are heated (and cooled) mechanically, where heat (cold) is transferred (lost) to the surroundings via ground floors and facades. The same happens via the exhaust of ventilation systems. In the last 5-10 years, UTCI (universal thermal comfort index) has been applied opening a potential to incorporate Ecospace position-information together with UTCI.

Ecospace expansion (built): It might be a hypothesis that buildings will always cause a reduction in Ecospace expansion. However, the landscaping around buildings has an impact on Ecospace. Some buildings have gardens and green spaces as part of their cohesive environment, which can also be the case on green facades, balconies and roofs. Some buildings also have water as part of the design. The built environment is often characterized by the systematic removal of the basis for Ecospace expansion - leaves are swept up, trees are pruned, grass is mowed, the ground is paved with concrete, asphalt etc. However, tree pruning could also create new habitats in trees, ancient trees may survive longer in urban areas than production forests and hedges between allotments may serve as safe spaces with limited disturbances (Goulart, 2023). Continuity of unaltered built areas show indicators of being preferred by humans as well (Mouratidis, 2020). Hence, Ecospace expansion data has the potential to influence design decisions regarding landscape design, but also the way the landscape is maintained.

Ecospace continuity (built): Ecospace spatial continuity is linked to the built environment in that it is often the built environment that creates barriers to spatial continuity - in the form of fences, access facilities (roads, parking) etc. Buildings can also be so large that they pose obstacles to spatial continuity. Boundaries arise from property ownership, sector boundaries in local planning and various forms of infrastructure. Simultaneously, the exercise of mirroring Ecospace theory into the built environment points to the potential of improved spatial continuity in the built environment as an indicator for improved social efficacy of the built environment, also for the human species. This points to a potential applicability of Ecospace theory as a way to test architectural hypotheses about spatial indicators of social efficacy also for the human species. In continuation hereof, the *temporal* Ecospace *continuity* has potential to provide design strategies based on historical demolition patterns, but also for identification of spatial indicators that enhance the affluence of human habitation, sociality and well-being over time opening a hitherto unexplored potential to qualify the built environment on its capacity for life as implied in Ecospace. However,

Ecospace continuity perspectives are not part of current research or practice. Ecospace continuity could provide non-human arguments, but also integrated human and non-human 'beyond Anthropocene arguments' (Tsing, 2012), for preserving building facilities (besides avoided CO^2-emissions and biodiversity-destruction from raw material extraction). Ecospace temporal continuity also has the potential to test hypotheses and inform design decisions in detail concerning the way buildings are maintained and transformed. Preservation of buildings increases temporal continuity and is a viewpoint that is not currently included in research and practice considerations. The demolition of buildings and subsequent construction of new buildings of course also affects Ecospace expansion and position via e.g. construction and excavation work as described above.

Ecospace *uniqueness (built)*: Ecospace uniqueness can be affected by the built environment, whose infrastructure or buildings create very large barriers, creating a kind of isolated 'islands'. The way the terrain is paved and drained around buildings can create rock-rich, extremely dry and compacted soils. With temporal continuity over a very long time these 'extreme' conditions can provide Ecospace for unique species. The same goes for the particular geometry of buildings - for example, very tall buildings, or at a microscale level in the detailing of relations between private and public space for example. From the point of view of architecture, the notion of Ecospace uniqueness, simultaneously provide a new spatial entry to the study of how specific unique qualities in architecture affect the social capacity for life for human and non-human species alike.

2.4 *Formulation and testing of interdisciplinary hypotheses*

Based on this mirroring of Ecospace into built environment to include human sociology in the Ecospace ecology, the main research question is further emphasized (is it possible to identify 'places' or situations where abstaining from over-consumption and excessive building would hold also experiences of affluences for lifeforms including humans?). Aesthetics, human wellbeing and the creative affluence in the living are inquired as entangled. The main research question asks whether a biodiverse neighborhood, celebrating creativity in nature can replace overconsumption, and constitute another kind of affluence? In Ecospace, low to intermediate resource levels (e.g. nutrients) give the highest number of species because in those communities there is a preponderance of frugal species that do not compete with each other (Grime 1973, Huston 1999). Is there a relation to architectural traits that celebrate modesty; thrive through sufficiency? Adding to this, a hypothesis was formulated; Temporal continuity of biotopes promotes both human well-being and biodiversity of greenspaces. Spatial and temporal continuity is one of the most important parameters for biodiversity both concerning species richness and for the conservation of unique species. Reusing and transforming existing structures is a basic sustainability and circularity strategy in the built environment. The question is whether this is more than a simple correlation? The proposed mirroring and integration of the human specia and the built environment into Ecospace theory opens a potential to combine and triangulate quantitative and qualitive measures across the three fields of knowledge; across interspecies diversity, aesthetics and well-being in this matter.

3 CONCLUSION

Based on one year of systematic interviews and collaboration across architecture, sociology and biology a foundation for mirroring concepts from Ecospace theory into the other two disciplines was laid. The choice of Ecospace as the foundation revealed a common desire to develop methods that 'communicate' with non-humans. Using Ecospace as a starting point challenged the methods of sociologists and architects. Because there is more built environment than biomaterial on the planet, methods in biology are also fundamentally challenged, and a deeper understanding of the ideas behind the design of an environment is necessary to further develop a continuous Ecospace framework in which terrestrial and built are conceivable as interrelated modes of a continuous transformation. By mirroring concepts from the built environment into Ecospace, a first step forward from the interviews was achieved. Two elements of meaning emerged; 1. That affluence in

the living world as expressed through its creativity 'is' present in the world in the same way as aesthetics in architecture and ethics in human relations. 2. That there is very little data collected on possible correlations between human and non-human affluence in low environmental-impact-environments. In conclusion, we have found that a rather slow and deep process of uncovering the ontology of each of the three disciplines is necessary but also rewarding and that this process itself becomes important. This is also a motivation to layout these process results for a larger group, to help calibrate the methods and processes in a conference setting.

4 FUTURE WORK

As mentioned in the introduction a motivation for the work is to formulate an interdisciplinary framework for the case studies as called for by leading post-growth theorists (Kallis et al. 2025). From these initial findings, ideas were developed for a series of case studies of buildings and their immediate surroundings in the scale of 40×40 m mirroring the scale investigated in the original terrestrial Ecospace pilot studies and inquired with the same rigor, testing a series of interspecies-wellbeing and affluence-related hypotheses. The cases involve biological studies of bio- and human diversity within the local spatial niche, architectural study of the materiality and spatial capacity of the local spatial niche and its footprint, and sociological study of the 'sense of affluence' across bio- and human sociality and its ethical implications, as a way to situate knowledge (Haraway 2020, Tsing, 2012). The research institutions involved have access to unique data on biodiversity, soil, buildings and citizens which will provide high-quality data to test the hypotheses. The question however remains, if the results of the mirroring and interviewing process is enough to create these case studies or if some further tools for reaching a transdisciplinary level has to come into play. Based on the inquiries into disciplinary methods an intermediate layer was identified where data visualization is hypothesized to perform as steppingstones towards transformative practices. All disciplines have visual forms of analysis that could be tested as synthesizing 'exercises' performed by researchers grounded in disciplinary knowledge from each of the three disciplines; 'Visual thick descriptions', 'Architectural Analytiques', 'visualizations of biotopes (Pink 2008, Frascari 1984, Nassar 2022). In this way, development of a transdisciplinary visual language prepares for interventions situating empirical science in local actions with the inhabitants.

4.1 *Studies and transformation of continuous biotopes*

Lastly it is also a question for future work to get from 'Ecospace' to 'Ecospacing' (Lury,2018). Each discipline holds action-oriented research designs coupled with an intervention repertoire and post-intervention evaluation methods exposing paths towards a regenerative practice. Biologists' interventions include restoration of habitats such as grasslands, ponds, introducing new habitas e.g. large rocks and dead wood, and including construction of new ecosystems such as green roofs and brown soils. Architectural research interventions establish passages between inside and outside, adjusting a tectonic expression (introducing a new material vocabulary), canopies, screens, new paths, connections and meeting places. Sociological interventions create "mini-publics", urban learning machines that move to engage with others and participate in an activity 'beyond discipline'. An ideal invention/prototyping process could include participatory design processes facilitated by sociologists. Draft designs of interventions are continuously visualized by architects and eventually produced in collaboration with users (Falanga 2020, Klaus & Kiehl 2021). The continuous data collection after occupancy, where the empirical case studies are baseline add information on biodiversity, footprint and perception of well-being, and how/if the creation process itself has changed views on affluence and interspecies relations.

REFERENCES

Angelo, H. & Greenberg, M. 2023. Environmentalizing Urban Sociology, *City & Community*, 22(4).

Brunbjerg A.K., Bruun H.H., Moeslund J.E., Sadler J., Svenning J. & Ejrnæs R. 2017. Ecospace: A unified framework for understanding variation in terrestrial biodiversity. *Basic and Applied Ecology* 18:86–94.
Elhacham, E., Ben-Uri, L., Grozovski, J., Bar-On, Y.M. & Milo, R. 2020. Global human-made mass exceeds all living biomass. *Nature* 588:442–444.
Falanga, R. 2020. Participatory design: participatory urban management. In *Sustainable Cities and Communities*. Springer International Publishing, 449–457.
Frascari, M. 1984 The tell-the-tail-detail. VIA 7.
Goulart, F. & Chappell M. 2023. Sparing or expanding? *Biodiversity and Conservation* 20(32): 1089–1104
Grime, J.P. 1973. Competitive exclusion in herbaceous vegetation. *Nature* 242:344–347.
Haraway, D. 1988. Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective. *Feminist Studies*, 14(3): 575.
Hauschild M.(ed.) 2018. *Life Cycle Assessment: Theory and Practice*. Springer International Publishing.
Huston, M.A. 1999. Local processes and regional patterns, *Oikos* 86: 393–401.
Hvejsel, M.F., Jensen, L.M.B., Albertsen, N., Ejrnæs, R. & Brunbjerg, A.K. 2024. 'Ecospaceing?' –
Towards a local framework for interspecies well-being and regeneration' in Fetzer, E (Ed) (2024) *TELOS Landscape Economy. Université Libre de Bruxelles* (forthcoming).
Jackson, T. 2017. *Prosperity without growth: foundations for the economy of tomorrow*, London, T&F.
Jakobsen, A.L., Jørgensen, A., Tølbøll, L., & Johnsen, S.B. (2022). Opening the black box of the relationship between neighborhood socioeconomic status and mental health: Neighborhood social-interactive characteristics as contextual mechanisms. *Health & Place*, 77, 102905. https://doi.org/10.1016/j.healthplace.2022.102905.
Kallis, G. Hickel, J., O'Neill, D.W., Jackson, T., Victor, P.A., Raworth, K., Schor, J.B., Steinberger, J. K., & Ürge-Vorsatz, D. 2025. Post-growth: the science of wellbeing within planetary boundaries. *The Lancet Planetary Health*, 9(1), e62–e78. https://doi.org/10.1016/S2542-5196(24)00310-3.
Klaus, V.H. & Kiehl, K. 2021. A conceptual framework for urban ecological restoration and rehabilitation. *Basic and Applied Ecology*, *52*: 82–94.
Leibold M.A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J.M., Hoopes, M.F., Holt, R.D., Shurin, J.B., Law, R., Tilman, D., Loreau, M. & Gonzalez, A. 2004. The metacommunity concept: a framework for multi-scale community ecology. *Ecology Letters* 7:601–613.
Lund, R.L. (2019). Dissecting the local : Territorial Scale and the Social Mechanisms of Place. Aalborg University.
Lund, R.L., Jørgensen, A., & Riis, O.P. (2024). Same-same, but different: A mixed-methods approach to enhance the understanding of spatial social science and the impact of neighborhoods. Geoforum Perspektiv, 22(42). https://doi.org/10.54337/ojs.perspektiv.v22i42.7098.
Lury, C. 2018. Activating the present of interdisciplinary methods, In C. Lury (ed.) Handbook of Interdisciplinary Research Methods. 1 edn Routledge pp. 1–23.
Lyle, J.T. 1994 *Regenerative Design for Sustainable Development*, Wiley NYC.
Macarthur R, and Wilson E. (1967) *The theory of island biogeography*. Princeton University Press, Princeton, New Jersey.
Meadows, D.H. et al. 2004. *The Limits to Growth: The 30-year Update*. Chelsea Green Pub. Co.
Mouratidis, K & Ramzi, H. 2020. Contemporary versus traditional styles in architecture and public space: A virtual reality study with 360-degree videos. *Cities* 97.
Nassar, D. 2022. Organism and Environment: The Aesthetic Foundations of Humboldt's Ecological Insight *Romantic Empiricism: Nature, Art, and Ecology from Herder to Humboldt*. Oxford.
Neumann, B. & Nünning, A. 2012. *Travelling Concepts for the Study of Culture*. De Gruyter.
Park, Robert Ezra (1952): Human Communities – The City and Human Ecology. Glencoe III: Free Press
Park, R.E., and E.W. Burgess (1925). *The City: Suggestions for Investigation of Human Behavior in the Urban Environment*. Chicago: University of Chicago Press.
Pink, S. 2008. Mobilising Visual Ethnography: Making Routes, Making Place and Making Images. *Forum Qualitative Social Research*, 9(3).
Sampson, R.J. (2011). Neighborhood effects, causal mechanisms and the social structure of the city. Analytical Sociology and Social Mechanisms, Sampson 2010, 227–249. https://doi.org/10.1017/CBO9780511921315.012.
Sampson, R.J. (2012). Great American city - Chicago and the enduring neighborhood effect [Book]. University of Chicago Press.Tsing, A.L. 2012. On nonscalability: The living world is not amenable to precision-nested scales. *Common knowledge*, *18*(3): 505–524.
UNEP 2023. global_status_report_buildings_construction_2023.pdf.
Watts, E., Hoßfeld, U. & Levit, G.S. 2019. Ecology and evolution: Haeckel's Darwinian paradigm. Trends in ecology & evolution 34(8): 681–683.